

task-8-syscall-audit

BOB-127 — Task 8 syscall audit (reproduced closure evidence)

Revision: 1 Last modified: 2026-09-02T00:00:00Z

Why this file exists

BOB-127 closed on 2026-08-19 (item_history id=172, event=Fixed, by=AI) citing evidence at .superpowers/sdd/task-8-syscall-audit.md. That path was **never committed** — git log --all -- '.superpowers/sdd/task-8-syscall-audit.md' returns nothing and .superpowers/ does not exist in the working tree — so the closure's captured proof was not producible on demand and workable-items validate refused it under §11.4.5 / §11.4.69 / §11.4.123 / §11.4.226.

The claim itself is **re-producible today**, so this file re-executes the audit and lands it at the project's canonical, tracked evidence location (docs/qa/<ATM-ID>/, §11.4.83 / §11.4.215) rather than an untracked scratch directory. The item_history evidence_path is corrected to point here.

Why the original vanished — known mechanism, not a new discovery. This project's own QA discovery ledger already records the class: entry SCRATCH-LOSS-2026-08-18 (docs/QA_DISCOVERY_LEDGER.md) documents .superpowers/sdd/ deliverables that were declared but never written, root-caused to producer subagents crashing on session rate limits mid-deliverable (§11.4.147(e)), with the follow-up **BOB-107** filed for a pre-dispatch precondition check. .superpowers/ is an untracked agent scratchpad, so anything cited from it is outside version control and unproducible the moment that scratchpad is gone.

Citing a scratchpad path as a closure's captured proof is the underlying defect; landing evidence under tracked docs/qa/ is the fix, and matches every other evidence row in this DB.

What BOB-127 was

Follow-up to the BOB-126 host-safety sweep (§11.4.263). The Task-8 audit found **2 test cases** in tests/unit/merge_service/test_public_tracker_subprocess_timeout.py that set an explicit integer mock.pid (12345, 1111) — satisfying the *production* BOB-126 pid > 1 invariant — but did **not** patch os.killpg / os.getpgid, so the real syscalls fired against hardcoded, non-owned PIDs. Low collision probability on a typical host, but a §11.4.263(C) test-side hygiene violation in the exact file authored to regression-guard against host-wide kills.

Fix: commit 8bedc5a (“test(BOB-126): patch os.killpg/getpgid in remaining 2 subprocess-timeout tests”) added patch.object(_search.os, "getpgid", ...) + patch.object(_search.os, "killpg") to both tests, copying the sanctioned pattern already used by the sibling test_process_group_kill_called_on_deadline.

Layer 1 — the fix is present at HEAD (60729c9)

```
$ grep -n "getpgid\|killpg\|def test_" tests/unit/merge_service/
test_public_tracker_subprocess_timeout.py
15:1. Process-group kill (`start_new_session=True` + `os.killpg`)
so
62:async def test_stuck_subprocess_killed_and_abandoned() -> None:
83:     patch.object(_search.os, "getpgid",
return_value=12345),
84:     patch.object(_search.os, "killpg"),
95:async def test_process_group_kill_called_on_deadline() -> None:
117:     patch.object(_search.os, "getpgid", return_value=9999)
as mock_getpgid,
118:     patch.object(_search.os, "killpg") as mock_killpg,
122:     mock_getpgid.assert_called_once_with(9999)
123:     mock_killpg.assert_called_once_with(9999,
pytest.approx(9)) # SIGKILL = 9
126:async def test_cleanup_timeout_abandons_zombie() -> None:
150:     patch.object(_search.os, "getpgid",
return_value=1111),
151:     patch.object(_search.os, "killpg"),
```

Both formerly-unguarded tests (lines 62 and 126) now carry the guard pair at lines 83-84 and 150-151.

Layer 2 — the 6 tests still pass (runtime)

Full log: `pytest_6_tests.log`

```
tests/.../test_stuck_subprocess_killed_and_abandoned      PASSED
[ 16%]
tests/.../test_process_group_kill_called_on_deadline      PASSED
[ 33%]
tests/.../test_cleanup_timeout_abandons_zombie            PASSED
[ 50%]
tests/.../test_outer_backstop_timeout_in_search_tracker   PASSED
[ 66%]
tests/.../test_return_exceptions_in_run_search            PASSED
[ 83%]
tests/.../test_return_exceptions_preserves_successful_tracker
PASSED [100%]
```

```
===== 6 passed in 20.63s
=====
```

(The closing commit claimed “All 6 tests in file still PASS (21.6s)”; re-measured today at 20.63 s — the same 6/6.)

Layer 3 — the audit is now mechanized, and it is GREEN

The audit’s finding is no longer a one-off human read: it is the standing pre-build gate CM-TEST-MOCK-PID-PATCHED-WHEN-REAL-PID (`scripts/pre_build/check_cm_test_mock_pid_patched_when_real_pid.sh`), which the BOB-127 body recorded as a recommended follow-up and which has since shipped.

GREEN — full log: `gate_cm_test_mock_pid_GREEN.log`

```
$ bash scripts/pre_build/
check_cm_test_mock_pid_patched_when_real_pid.sh
check_cm_test_mock_pid_patched_when_real_pid - CM-TEST-MOCK-PID-
PATCHED-WHEN-REAL-PID static pre-build gate
  scope: default (tests, excluding tests/pre_build/), repo-root: /
home/milosvasic/Projects/boba
  forward window: 20 lines; os.killpg-exemption window: ±30 lines
  files scanned: 336
```

```
PASS: CM-TEST-MOCK-PID-PATCHED-WHEN-REAL-PID clean across 336
file(s)
GATE_EXIT=0
```

Layer 4 — §1.1 paired mutation: the gate genuinely catches BOB-127's defect

RED — full log: `gate_cm_test_mock_pid_RED_mutation.log`

Mutation: delete lines 83-84 (the two guard lines) from `test_stuck_subprocess_killed_and_abandoned`, restoring exactly the pre-8bedc5a defect. The gate is a **static source scan**, so this RED was captured **without ever firing an unguarded killpg** against a non-owned PID (§11.4.263 — the defect is proven by detection, never by re-executing the dangerous syscall).

=== FINDINGS ===

```
FAIL: .../test_public_tracker_subprocess_timeout.py:70: real-pid
subprocess-mock with
NO os.killpg-targeting patch – proc = AsyncMock()
[identifier=proc] (has `proc.pid =
<int>` but no `os.killpg`-targeting patch within 30 lines; see
test_process_group_kill_called_on_deadline ... for the sanctioned
fix pattern)
```

```
FAIL: 1 unpatched real-pid test-subprocess-mock hit(s) (CM-TEST-
MOCK-PID-PATCHED-WHEN-REAL-PID, BOB-127)
GATE_EXIT_UNDER_MUTATION=1
```

The gate names **BOB-127** in its own failure text. The mutation was reverted immediately and the working tree re-verified clean (`git diff --stat` empty for that file) before any further work — §11.4.84 working-tree quiescence.

Polarity pair: RED exit 1 on the defect present → GREEN exit 0 on the defect fixed. The guard is therefore validated instrumentation, not decoration (§11.4.115(F)).

Verdict

BOB-127's closure claim is **re-produced and true at HEAD 60729c9**: the defect is fixed in source, the 6 tests pass, the audit is mechanized as a standing gate that is GREEN, and that gate provably FAILs when the defect is reintroduced. The closure stands; only its `evidence_path` was wrong, and it now points at this file.

Honest boundary (§11.4.6)

- The **original** `.superpowers/sdd/task-8-syscall-audit.md` is gone and was never committed; its verbatim prose is **not** recovered. What is reproduced here is the audit's *substance* — the finding, the fix, and machine-verifiable proof of both — re-derived today from the tracked commit 8bedc5a and the live tree.
- This evidence covers BOB-127's specific finding (2 unguarded test mocks in one file). It is not a claim that the whole test corpus is free of every host-safety anti-pattern; the standing gate's 336-file scan is the ongoing instrument for that.